Supplementary Information

Search strategy

For the selected databases, the following search string was developed, which was combined with two concepts and keyword terms: - concept 1 = "stroke"; (cerebrovascular disorders/basal ganglia cerebrovascular disease/ brain ischaemia, or carotid artery disease/ or intracranial arterial disease/ or intracranial embolism and thrombosis/ or intracranial haemorrhages/ or stroke/ or brain infarction/ or vertebral artery dissection), OR (stroke or cerebrovascular or brain vascular or cerebral vascular, or cva orapoplex), OR (brain or cerebral or vertebrobasilar or hemisphere or intracranial or intracerebral or infratentorial or supratentorial or MCA or anterior circulation or posterior circulation or basal ganglia) and (ischaemic, or infarct, or thrombosis, or embolism), OR (brain or cerebral or intracerebral or intracranial or parenchymal or intraventricular or infratentorial or supratentorial or basal ganglia) and (haemorrhage or hemorrhage or haematoma or hematoma or bleed), OR (stroke or poststroke or post*stroke or cerebrovascular or brain vascular or cerebral vascular or cva or apoplexy or ischaemic attack or ischemic attack or tia or neurological deficit or SAH or AVM). Concept 2 – "microRNA"; (microRNA/ micro RNA/ miR or miRNA or mir), OR non-coding, OR (primary microRNA, primary miRNA, or pre-miRNA, or pri-miRNA), OR (small temporal RNA or stRNA). Search results were concept 1 AND concept 2 combined and limited to publications after 1993 until present (October 2021, inclusively).

Inclusion and Exclusion Criteria

As prespecified in the protocol (http://syrf.org.uk/protocols/ - published 22.8.17).

Inclusion criteria: - any experiments profiling microRNA expression in humans or animals with stroke; in animal models and patients of all ages and both sexes; with appropriate use of control non-stroke patients/animals with comorbidities; microRNA expression, where quantitative measure of microRNA expression is reported, where microRNA expression outcome data has then been quantitatively measured by quantitative reverse transcription-polymerase chain reaction (qRT-PCR); studies published in peer reviewed scientific journals; in all languages were included. As long as microRNA data could be extracted, studies were still included if a subset of the included participants/experimental groups met the inclusion criteria. MicroRNA expression and any summary quantitative measure of microRNA expression, where an estimate of precision had been reported; microRNA expression data which has been quantitatively measured by some form of qRT-PCR – as opposed to non-validated array data.

Exclusion criteria: - non-stroke studies (such as traumatic brain injury or subarachnoid haemorrhage), intervention trials that may influence microRNA expression, reviews, protocol papers, editorials; studies which do not report quantitative measure of microRNA expression, and do not validate their data with qRT-PCR; studies which do not use appropriate non-stroke controls; publications published before 1993 and not published in peer reviewed journals.

Data extraction

Details of interest included: - journal title, year of publication, animal model (sex, species, strain), stroke model or stroke subtype, functional outcome measure(s) (primary and secondary), methods of microRNA quantification, and expression levels of circulating microRNAs.

CAMARADES checklist

Peer reviewed publication, use of control non-stroke patients or animals with comorbidities, samples size calculation, statement of compliance with ethical requirements, statement of possible conflicts of interest, pre-specified inclusion, and exclusion criteria, reporting of patient/animals excluded from analysis, source of study funding reported. Specifically, for non-human studies, the following evaluations were made: - control of temperature, randomised group allocation/blinded induction of stroke, blinded assessment of outcome, statement of compliance with animal welfare requirements, and monitoring of physiological variables throughout the study.

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